



DHA SUFFA UNIVERSITY

Off Khayaban-e-Tufail, Phase -VII (Extension), DHA, Karachi - 75500

Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN CYBER SECURITY

BS-CYS-2020

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1301	Calculus and Analytical Geometry	3
2	CYS-1001	Programming Fundamentals	3
3	CYS-1003	Discrete Structures	3
4	CYS-1201	Application of Information and Communication Technologies	2
5	CYS1001L	Programming Fundamentals Lab	1
6	CYS1201L	Application of Information & Communication Technologies Lab	1
7	HU-1002	Functional English	3
Total			16

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	BS-1402	Probability and Statistics	3
3	CYS-1002	Object Oriented Programming	3
4	CYS-1004	Database Systems	3
5	CYS1002L	Object Oriented Programming Lab	1
6	CYS1004L	Database Systems Lab	1
7	HU-1001	Communication & Presentation Skills	3
Total			17

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-2302	Differential Equations	3
2	CYS-2001	Data Structures & Algorithms	3
3	CYS-2002	Information Security	3
4	CYS-2004	Artificial Intelligence	3
5	CYS-2101	Digital Logic Design	3
6	CYS2001L	Data Structures & Algorithms Lab	1
7	CYS2004L	Artificial Intelligence Lab	1
8	CYS2101L	Digital Logic Design Lab	1
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	CYS-2003	Computer Networks	3
2	CYS-2005	Computer Organization & Assembly Language	3
3	CYS-2102	Introduction to Cyber Security	3
4	CYS-2103	Analysis of Algorithms	3
5	CYS-22XX	CySec Elective-1	3
6	CYS2003L	Computer Networks Lab	1
7	CYS2005L	Computer Organization & Assembly Language Lab	1
Total			17

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3001	Operating System	3
2	CYS-3100	Network Security	2
3	CYS-3104	Information Assurance	3
4	CYS-3201	Digital Forensics	2
5	CYS-33XX	CySec Elective-2	3
6	CYS3001L	Operating System Lab	1
7	CYS3100L	Network Security Lab	1
8	CYS3201L	Digital Forensics Lab	1
9	UE-XXXX	Uni Elective-1	3
Total			19

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3103	Parallel & Distributed Computing	2
2	CYS-33XX	CySec Elective-3	3
3	CYS-33XX	CySec Elective-4	3
4	CYS-3403	Secure Software Design and Development	2
5	CYS-3503	Vulnerability Assessment & Reverse Engineering	2
6	CYS3103L	Parallel & Distributed Computing Lab	1
7	CYS3403L	Secure Software Design and Development Lab	1
8	CYS3503L	Vulnerability Assessment & Reverse Engineering Lab	1
9	UE-XXXX	Uni-Elective-2	3
Total			18

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CYS-1005	Software Engineering	3
2	CYS-4100	Final Year Project-I	2
3	HU-2001	Technical & Business Writing	3
4	HU-4001	Islamic Studies & Ethics	2
5	UE-XXXX	Uni Elective-3	3
Total			13

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CYS-4102	Final Year Project-II	4
2	HU-2101	Pakistan Studies	2
3	HU-4201	Professional Practices	3
4	UE-XXXX	Uni Elective-4	3
Total			12

Total Credit Hours Required: 130



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN CYBER SECURITY

BS-CYS-2020 (Minorities)

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1301	Calculus and Analytical Geometry	3
2	CYS-1001	Programming Fundamentals	3
3	CYS-1003	Discrete Structures	3
4	CYS-1201	Application of Information and Communication Technologies	2
5	CYS1001L	Programming Fundamentals Lab	1
6	CYS1201L	Application of Information & Communication Technologies Lab	1
7	HU-1002	Functional English	3
Total			16

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	BS-1402	Probability and Statistics	3
3	CYS-1002	Object Oriented Programming	3
4	CYS-1004	Database Systems	3
5	CYS1002L	Object Oriented Programming Lab	1
6	CYS1004L	Database Systems Lab	1
7	HU-1001	Communication & Presentation Skills	3
Total			17

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-2302	Differential Equations	3
2	CYS-2001	Data Structures & Algorithms	3
3	CYS-2002	Information Security	3
4	CYS-2004	Artificial Intelligence	3
5	CYS-2101	Digital Logic Design	3
6	CYS2001L	Data Structures & Algorithms Lab	1
7	CYS2004L	Artificial Intelligence Lab	1
8	CYS2101L	Digital Logic Design Lab	1
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	CYS-2003	Computer Networks	3
2	CYS-2005	Computer Organization & Assembly Language	3
3	CYS-2102	Introduction to Cyber Security	3
4	CYS-2103	Analysis of Algorithms	3
5	CYS-22XX	CySec Elective-1	3
6	CYS2003L	Computer Networks Lab	1
7	CYS2005L	Computer Organization & Assembly Language Lab	1
Total			17

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3001	Operating System	3
2	CYS-3100	Network Security	2
3	CYS-3104	Information Assurance	3
4	CYS-3201	Digital Forensics	2
5	CYS-33XX	CySec Elective-2	3
6	CYS3001L	Operating System Lab	1
7	CYS3100L	Network Security Lab	1
8	CYS3201L	Digital Forensics Lab	1
9	UE-XXXX	Uni Elective-1	3
Total			19

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3103	Parallel & Distributed Computing	2
2	CYS-33XX	CySec Elective-3	3
3	CYS-33XX	CySec Elective-4	3
4	CYS-3403	Secure Software Design and Development	2
5	CYS-3503	Vulnerability Assessment & Reverse Engineering	2
6	CYS3103L	Parallel & Distributed Computing Lab	1
7	CYS3403L	Secure Software Design and Development Lab	1
8	CYS3503L	Vulnerability Assessment & Reverse Engineering Lab	1
9	UE-XXXX	Uni-Elective-2	3
Total			18

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CYS-1005	Software Engineering	3
2	CYS-4100	Final Year Project-I	2
3	HU-2001	Technical & Business Writing	3
4	HU-4001	Islamic Studies & Ethics	2
5	UE-XXXX	Uni Elective-3	3
Total			13

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CYS-4102	Final Year Project-II	4
2	HU-2101	Pakistan Studies	2
3	HU-4201	Professional Practices	3
4	UE-XXXX	Uni Elective-4	3
Total			12

Total Credit Hours Required: 130



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN CYBER SECURITY

BS-CYS-2020 (Minorities)

PreMed

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1002	Elementary Mathematics1	0
2	BS-1301	Calculus and Analytical Geometry	3
3	CYS-1001	Programming Fundamentals	3
4	CYS-1003	Discrete Structures	3
5	CYS-1201	Application of Information and Communication Technologies	2
6	CYS1001L	Programming Fundamentals Lab	1
7	CYS1201L	Application of Information & Communication Technologies Lab	1
8	HU-1002	Functional English	3
Total			16

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-2302	Differential Equations	3
2	CYS-2001	Data Structures & Algorithms	3
3	CYS-2002	Information Security	3
4	CYS-2004	Artificial Intelligence	3
5	CYS-2101	Digital Logic Design	3
6	CYS2001L	Data Structures & Algorithms Lab	1
7	CYS2004L	Artificial Intelligence Lab	1
8	CYS2101L	Digital Logic Design Lab	1
Total			18

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3001	Operating System	3
2	CYS-3100	Network Security	2
3	CYS-3104	Information Assurance	3
4	CYS-3201	Digital Forensics	2
5	CYS-33XX	CySec Elective-2	3
6	CYS3001L	Operating System Lab	1
7	CYS3100L	Network Security Lab	1
8	CYS3201L	Digital Forensics Lab	1
9	UE-XXXX	Uni Elective-1	3
Total			19

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CYS-1005	Software Engineering	3
2	CYS-4100	Final Year Project-I	2
3	HU-2001	Technical & Business Writing	3
4	HU-4001	Islamic Studies & Ethics	2
5	UE-XXXX	Uni Elective-3	3
Total			13

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1003	Elementary Mathematics II	0
2	BS-1302	Linear Algebra	3
3	BS-1402	Probability and Statistics	3
4	CYS-1002	Object Oriented Programming	3
5	CYS-1004	Database Systems	3
6	CYS1002L	Object Oriented Programming Lab	1
7	CYS1004L	Database Systems Lab	1
8	HU-1001	Communication & Presentation Skills	3
Total			17

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	CYS-2003	Computer Networks	3
2	CYS-2005	Computer Organization & Assembly Language	3
3	CYS-2102	Introduction to Cyber Security	3
4	CYS-2103	Analysis of Algorithms	3
5	CYS-22XX	CySec Elective-1	3
6	CYS2003L	Computer Networks Lab	1
7	CYS2005L	Computer Organization & Assembly Language Lab	1
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3103	Parallel & Distributed Computing	2
2	CYS-33XX	CySec Elective-3	3
3	CYS-33XX	CySec Elective-4	3
4	CYS-3403	Secure Software Design and Development	2
5	CYS-3503	Vulnerability Assessment & Reverse Engineering	2
6	CYS3103L	Parallel & Distributed Computing Lab	1
7	CYS3403L	Secure Software Design and Development Lab	1
8	CYS3503L	Vulnerability Assessment & Reverse Engineering Lab	1
9	UE-XXXX	Uni-Elective-2	3
Total			18

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CYS-4102	Final Year Project-II	4
2	HU-2101	Pakistan Studies	2
3	HU-4201	Professional Practices	3
4	UE-XXXX	Uni Elective-4	3
Total			12

Total Credit Hours Required: 130



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PROGRAM: BACHELOR OF SCIENCE IN CYBER SECURITY

BS-CYS-2020 PreMed

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1002	Elementary Mathematics1	0
2	BS-1301	Calculus and Analytical Geometry	3
3	CYS-1001	Programming Fundamentals	3
4	CYS-1003	Discrete Structures	3
5	CYS-1201	Application of Information and Communication Technologies	2
6	CYS1001L	Programming Fundamentals Lab	1
7	CYS1201L	Application of Information & Communication Technologies Lab	1
8	HU-1002	Functional English	3
Total			16

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-2302	Differential Equations	3
2	CYS-2001	Data Structures & Algorithms	3
3	CYS-2002	Information Security	3
4	CYS-2004	Artificial Intelligence	3
5	CYS-2101	Digital Logic Design	3
6	CYS2001L	Data Structures & Algorithms Lab	1
7	CYS2004L	Artificial Intelligence Lab	1
8	CYS2101L	Digital Logic Design Lab	1
Total			18

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3001	Operating System	3
2	CYS-3100	Network Security	2
3	CYS-3104	Information Assurance	3
4	CYS-3201	Digital Forensics	2
5	CYS-33XX	CySec Elective-2	3
6	CYS3001L	Operating System Lab	1
7	CYS3100L	Network Security Lab	1
8	CYS3201L	Digital Forensics Lab	1
9	UE-XXXX	Uni Elective-1	3
Total			19

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CYS-1005	Software Engineering	3
2	CYS-4100	Final Year Project-I	2
3	HU-2001	Technical & Business Writing	3
4	HU-4001	Islamic Studies & Ethics	2
5	UE-XXXX	Uni Elective-3	3
Total			13

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1003	Elementary Mathematics II	0
2	BS-1302	Linear Algebra	3
3	BS-1402	Probability and Statistics	3
4	CYS-1002	Object Oriented Programming	3
5	CYS-1004	Database Systems	3
6	CYS1002L	Object Oriented Programming Lab	1
7	CYS1004L	Database Systems Lab	1
8	HU-1001	Communication & Presentation Skills	3
Total			17

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	CYS-2003	Computer Networks	3
2	CYS-2005	Computer Organization & Assembly Language	3
3	CYS-2102	Introduction to Cyber Security	3
4	CYS-2103	Analysis of Algorithms	3
5	CYS-22XX	CySec Elective-1	3
6	CYS2003L	Computer Networks Lab	1
7	CYS2005L	Computer Organization & Assembly Language Lab	1
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3103	Parallel & Distributed Computing	2
2	CYS-33XX	CySec Elective-3	3
3	CYS-33XX	CySec Elective-4	3
4	CYS-3403	Secure Software Design and Development	2
5	CYS-3503	Vulnerability Assessment & Reverse Engineering	2
6	CYS3103L	Parallel & Distributed Computing Lab	1
7	CYS3403L	Secure Software Design and Development Lab	1
8	CYS3503L	Vulnerability Assessment & Reverse Engineering Lab	1
9	UE-XXXX	Uni-Elective-2	3
Total			18

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CYS-4102	Final Year Project-II	4
2	HU-2101	Pakistan Studies	2
3	HU-4201	Professional Practices	3
4	UE-XXXX	Uni Elective-4	3
Total			12

Total Credit Hours Required: 130



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN CYBER SECURITY

BS-CYS-2024

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1101	Applied Physics	3
2	BS-1301	Calculus & Analytical Geometry	3
3	CS-1001	Application of Information and Communication Technologies	2
4	CS-1001L	Applications of Information & Communication Technologies LAB	1
5	CS-1002	Programming Fundamentals	3
6	CS-1002L	Programming Fundamentals Lab	1
7	HU-1001	Functional English	3
8	HU-2201	Islamic Studies	2
Total			18

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-2301	Multivariate Calculus	3
2	CS-2007	Data Structures & Algorithms	3
3	CS-2007L	Data Structures & Algorithms Lab	1
4	CS-2201	Computer Networks	3
5	CS-2201L	Computer Networks Lab	1
6	CS-2801	Software Engineering	3
7	MS-3106	Organizational Behaviour	2
Total			16

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	CS-3301	Artificial Intelligence	2
2	CS-3301L	Artificial Intelligence Lab	1
3	CYS-3001	Cyber Security	2
4	CYS-3XXX	CySec Elective-3	3
5	CYS-3XXX	CySec Elective-1	2
6	CYS-3XXX	CySec Elective-2	2
7	CYS3001L	Cyber Security Lab	1
8	CYS3XXXL	CySec Elective-2 Lab	1
9	CYS3XXXL	CySec Elective-1 Lab	1
Total			15

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-4012	Analysis of Algorithms	3
2	CS-4013	Final Year Project-I	3
3	CYS-3306	CySec Elective-5	3
4	CYS-3307	CySec Elective-6	3
5	CYS-3XXX	CySec Elective-4	3
6	HU-32XX	Fehm-e-Quran – II	1
7	SS-4001	Elective Supporting	3
Total			19

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	CS-1003	Discrete Structures	3
3	CS-1004	Digital Logic Design	2
4	CS-1004L	Digital Logic Design Lab	1
5	CS-1005	Object Oriented Programming	3
6	CS-1005L	Object Oriented Programming Lab	1
7	HU-1006	Pakistan Studies	2
8	HU-2203	Expository Writing	3
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	CS-2008	Computer Organization & Assembly Language	2
3	CS-2008L	Computer Organization & Assembly Language Lab	1
4	CS-2009	Operating Systems	3
5	CS-2009L	Operating Systems Lab	1
6	CS-2101	Database Systems	3
7	CS-2101L	Database Systems Lab	1
8	CS-2202	Information Security	2
9	CS-2202L	Information Security Lab	1
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3002	Parallel & Distributed Computing	2
2	CYS-3201	Information Assurance	2
3	CYS-3301	Network Security	2
4	CYS-3401	Secure Software Design and Development	2
5	CYS-3503	Digital Forensics	2
6	CYS2301L	Information Assurance Lab	1
7	CYS3002L	Parallel & Distributed Computing Lab	1
8	CYS3301L	Network Security Lab	1
9	CYS3401L	Secure Software Design and Development Lab	1
10	CYS3503L	Digital Forensics Lab	1
11	HU-321XX	Fehm-e-Quran – I	1
Total			16

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CS-4014	Final Year Project - II	3
2	HU-3041	Professional Practice	2
3	HU-4001	Technical & Business Writing	3
4	HU-4200	Civics and Community Engagement	2
5	HU-XXXX	University Elective-II	3
6	MS-XXXX	Entrepreneurship	2
Total			15

Total Credit Hours Required: 134



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN CYBER SECURITY

BS-CYS-2024 (Minorities)

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1101	Applied Physics	3
2	BS-1301	Calculus & Analytical Geometry	3
3	CS-1001	Application of Information and Communication Technologies	2
4	CS-1001L	Applications of Information & Communication Technologies LAB	1
5	CS-1002	Programming Fundamentals	3
6	CS-1002L	Programming Fundamentals Lab	1
7	HU-1001	Functional English	3
8	HU-1603	Ethics	2
Total			18

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-2301	Multivariate Calculus	3
2	CS-2007	Data Structures & Algorithms	3
3	CS-2007L	Data Structures & Algorithms Lab	1
4	CS-2201	Computer Networks	3
5	CS-2201L	Computer Networks Lab	1
6	CS-2801	Software Engineering	3
7	MS-3106	Organizational Behaviour	2
Total			16

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	CS-3301	Artificial Intelligence	2
2	CS-3301L	Artificial Intelligence Lab	1
3	CYS-3001	Cyber Security	2
4	CYS-3XXX	CySec Elective-1	2
5	CYS-3XXX	CySec Elective-3	3
6	CYS-3XXX	CySec Elective-2	2
7	CYS3001L	Cyber Security Lab	1
8	CYS3XXXL	CySec Elective-2 Lab	1
9	CYS3XXXL	CySec Elective-1 Lab	1
Total			15

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-4012	Analysis of Algorithms	3
2	CS-4013	Final Year Project-I	3
3	CYS-3306	CySec Elective-5	3
4	CYS-3307	CySec Elective-6	3
5	CYS-3XXX	CySec Elective-4	3
6	SS-4001	Elective Supporting	3
Total			18

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	CS-1003	Discrete Structures	3
3	CS-1004	Digital Logic Design	2
4	CS-1004L	Digital Logic Design Lab	1
5	CS-1005	Object Oriented Programming	3
6	CS-1005L	Object Oriented Programming Lab	1
7	HU-1006	Pakistan Studies	2
8	HU-2203	Expository Writing	3
Total			18

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	CS-2008	Computer Organization & Assembly Language	2
3	CS-2008L	Computer Organization & Assembly Language Lab	1
4	CS-2009	Operating Systems	3
5	CS-2009L	Operating Systems Lab	1
6	CS-2010	Information Security	2
7	CS-2010L	Information Security Lab	1
8	CS-2101	Database Systems	3
9	CS-2101L	Database Systems Lab	1
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3002	Parallel & Distributed Computing	2
2	CYS-3201	Information Assurance	2
3	CYS-3301	Network Security	2
4	CYS-3401	Secure Software Design and Development	2
5	CYS-3503	Digital Forensics	2
6	CYS2301L	Information Assurance Lab	1
7	CYS3002L	Parallel & Distributed Computing Lab	1
8	CYS3301L	Network Security Lab	1
9	CYS3401L	Secure Software Design and Development Lab	1
10	CYS3503L	Digital Forensics Lab	1
11	HU-1425	Religion, Culture and Society	2
Total			17

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CS-4014	Final Year Project - II	3
2	HU-3041	Professional Practice	2
3	HU-4001	Technical & Business Writing	3
4	HU-4200	Civics and Community Engagement	2
5	HU-XXXX	University Elective-II	3
6	MS-XXXX	Entrepreneurship	2
Total			15

Total Credit Hours Required: 134



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Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN CYBER SECURITY

BS-CYS-2024 (Minorities)

PreMed

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1002	Elementary Mathematics1	0
2	BS-1101	Applied Physics	3
3	CS-1001	Application of Information and Communication Technologies	2
4	CS-1001L	Applications of Information & Communication Technologies LAB	1
5	CS-1002	Programming Fundamentals	3
6	CS-1002L	Programming Fundamentals Lab	1
7	HU-1001	Functional English	3
8	HU-1603	Ethics	2
Total			15

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1003	Elementary Mathematics II	0
2	BS-1301	Calculus and Analytical Geometry	3
3	CS-1004	Digital Logic Design	2
4	CS-1004L	Digital Logic Design Lab	1
5	CS-1005	Object Oriented Programming	3
6	CS-1005L	Object Oriented Programming Lab	1
7	HU-1006	Pakistan Studies	2
8	HU-2203	Expository Writing	3
Total			15

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	CS-1003	Discrete Structures	3
3	CS-2007	Data Structures & Algorithms	3
4	CS-2007L	Data Structures & Algorithms Lab	1
5	CS-2201	Computer Networks	3
6	CS-2201L	Computer Networks Lab	1
7	CS-2801	Software Engineering	3
8	MS-3106	Organizational Behaviour	2
Total			19

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-2301	Multivariate Calculus	3
2	CS-2008	Computer Organization & Assembly Language	2
3	CS-2008L	Computer Organization & Assembly Language Lab	1
4	CS-2009	Operating Systems	3
5	CS-2009L	Operating Systems Lab	1
6	CS-2010	Information Security	2
7	CS-2010L	Information Security Lab	1
8	CS-2101	Database Systems	3
9	CS-2101L	Database Systems Lab	1
Total			17

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	CS-3301	Artificial Intelligence	2
3	CS-3301L	Artificial Intelligence Lab	1
4	CYS-3001	Cyber Security	2
5	CYS-3XXX	CySec Elective-2	2
6	CYS-3XXX	CySec Elective-1	2
7	CYS-3XXX	CySec Elective-3	3
8	CYS3001L	Cyber Security Lab	1
9	CYS3XXXL	CySec Elective-1 Lab	1
10	CYS3XXXL	CySec Elective-2 Lab	1
Total			18

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3002	Parallel & Distributed Computing	2
2	CYS-3201	Information Assurance	2
3	CYS-3301	Network Security	2
4	CYS-3401	Secure Software Design and Development	2
5	CYS-3503	Digital Forensics	2
6	CYS2301L	Information Assurance Lab	1
7	CYS3002L	Parallel & Distributed Computing Lab	1
8	CYS3301L	Network Security Lab	1
9	CYS3401L	Secure Software Design and Development Lab	1
10	CYS3503L	Digital Forensics Lab	1
11	HU-1425	Religion, Culture and Society	2
Total			17

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-4012	Analysis of Algorithms	3
2	CS-4013	Final Year Project-I	3
3	CYS-3306	CySec Elective-5	3
4	CYS-3307	CySec Elective-6	3
5	CYS-3XXX	CySec Elective-4	3
6	SS-4001	Elective Supporting	3
Total			18

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CS-4014	Final Year Project - II	3
2	HU-3041	Professional Practices	2
3	HU-4001	Technical & Business Writing	3
4	HU-4200	Civics and Community Engagement	2
5	HU-XXXX	University Elective-II	3
6	MS-XXXX	Entrepreneurship	2
Total			15

Total Credit Hours Required: 134



DHA SUFFA UNIVERSITY

Off Khayaban-e-Tufail, Phase -VII (Extension), DHA, Karachi - 75500

Plan of Study

PROGRAM: BACHELOR OF SCIENCE IN CYBER SECURITY

BS-CYS-2024 PreMed

Semester 1

S. No.	Course Code	Course Title	Credit Hours
1	BS-1002	Elementary Mathematics1	0
2	BS-1101	Applied Physics	3
3	CS-1001	Application of Information and Communication Technologies	2
4	CS-1001L	Applications of Information & Communication Technologies LAB	1
5	CS-1002	Programming Fundamentals	3
6	CS-1002L	Programming Fundamentals Lab	1
7	HU-1001	Functional English	3
8	HU-2201	Islamic Studies	2
Total			15

Semester 3

S. No.	Course Code	Course Title	Credit Hours
1	BS-1302	Linear Algebra	3
2	CS-1003	Discrete Structures	3
3	CS-2007	Data Structures & Algorithms	3
4	CS-2007L	Data Structures & Algorithms Lab	1
5	CS-2201	Computer Networks	3
6	CS-2201L	Computer Networks Lab	1
7	CS-2801	Software Engineering	3
8	MS-3106	Organizational Behaviour	2
Total			19

Semester 5

S. No.	Course Code	Course Title	Credit Hours
1	BS-1402	Probability and Statistics	3
2	CS-3301	Artificial Intelligence	2
3	CS-3301L	Artificial Intelligence Lab	1
4	CYS-3001	Cyber Security	2
5	CYS-3XXX	CySec Elective-1	2
6	CYS-3XXX	CySec Elective-2	2
7	CYS-3XXX	CySec Elective-3	3
8	CYS3001L	Cyber Security Lab	1
9	CYS3XXXL	CySec Elective-2 Lab	1
10	CYS3XXXL	CySec Elective-1 Lab	1
Total			18

Semester 7

S. No.	Course Code	Course Title	Credit Hours
1	CS-4012	Analysis of Algorithms	3
2	CS-4013	Final Year Project-I	3
3	CYS-3306	CySec Elective-5	3
4	CYS-3307	CySec Elective-6	3
5	CYS-3XXX	CySec Elective-4	3
6	HU-32XX	Fehm-e-Quran – II	1
7	SS-4001	Elective Supporting	3
Total			19

Semester 2

S. No.	Course Code	Course Title	Credit Hours
1	BS-1003	Elementary Mathematics II	0
2	BS-1301	Calculus and Analytical Geometry	3
3	CS-1004	Digital Logic Design	2
4	CS-1004L	Digital Logic Design Lab	1
5	CS-1005	Object Oriented Programming	3
6	CS-1005L	Object Oriented Programming Lab	1
7	HU-1006	Pakistan Studies	2
8	HU-2203	Expository Writing	3
Total			15

Semester 4

S. No.	Course Code	Course Title	Credit Hours
1	BS-2301	Multivariate Calculus	3
2	CS-2008	Computer Organization & Assembly Language	2
3	CS-2008L	Computer Organization & Assembly Language Lab	1
4	CS-2009	Operating Systems	3
5	CS-2009L	Operating Systems Lab	1
6	CS-2010	Information Security	2
7	CS-2010L	Information Security Lab	1
8	CS-2101	Database Systems	3
9	CS-2101L	Database Systems Lab	1
Total			17

Semester 6

S. No.	Course Code	Course Title	Credit Hours
1	CYS-3002	Parallel & Distributed Computing	2
2	CYS-3201	Information Assurance	2
3	CYS-3301	Network Security	2
4	CYS-3401	Secure Software Design and Development	2
5	CYS-3503	Digital Forensics	2
6	CYS2301L	Information Assurance Lab	1
7	CYS3002L	Parallel & Distributed Computing Lab	1
8	CYS3301L	Network Security Lab	1
9	CYS3401L	Secure Software Design and Development Lab	1
10	CYS3503L	Digital Forensics Lab	1
11	HU-32XX	Fehm-e-Quran – I	1
Total			16

Semester 8

S. No.	Course Code	Course Title	Credit Hours
1	CS-4014	Final Year Project - II	3
2	HU-3041	Professional Practice	2
3	HU-4001	Technical & Business Writing	3
4	HU-4200	Civics and Community Engagement	2
5	HU-XXXX	University Elective-II	3
6	MS-XXXX	Entrepreneurship	2
Total			15

Total Credit Hours Required: 134